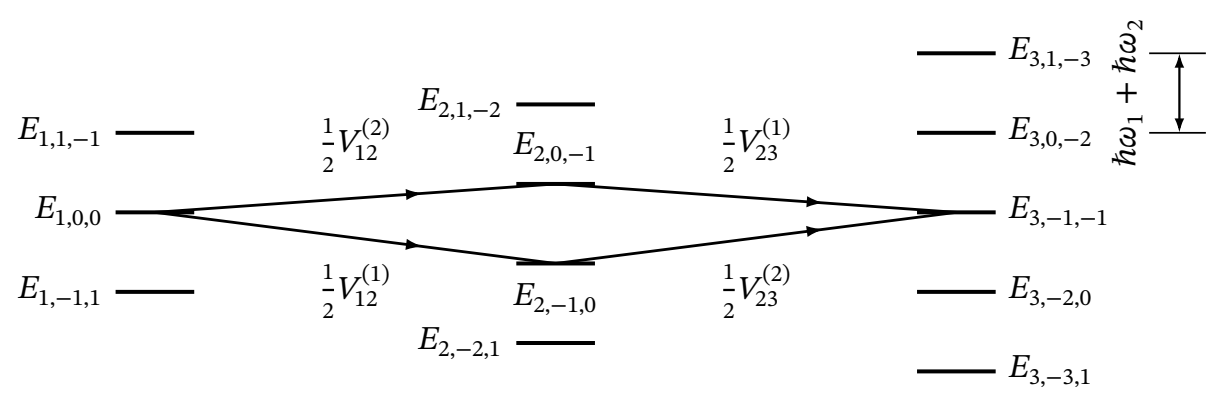
$E_{1,1,-1}$
$E_{1,0,0}$
$E_{1,-1,1}$
$\frac{1}{2}V_{12}^{(2)}$
$\frac{1}{2}V_{12}^{(1)}$
$E_{2,1,-2}$
$E_{2,0,-1}$
$E_{2,-1,0}$
$E_{2,-2,1}$
$\frac{1}{2}V_{23}^{(1)}$
$\frac{1}{2}V_{23}^{(2)}$
$E_{3,1,-3}$
$E_{3,0,-2}$
$E_{3,-1,-1}$
$E_{3,-2,0}$
$E_{3,-3,1}$
$\hbar\omega_1 + \hbar\omega_2$